TEXORA · LECTURE SHEET

Introduction to Nanoscience and Nanotechnology

Why matter behaves differently when it is small: the nanoscale, the surface-area effect, and the size dependence of mechanical, thermodynamic, electronic and catalytic properties — the physics underneath every nano claim in textiles.

TE-4206 · TECHNICAL TEXTILE AND NANO-TECHNOLOGY · TOPIC 6 · LEVEL 4 TERM 2

1. Definitions and scale

Nanoscience is the study of materials and phenomena at the scale of roughly **1–100 nanometres**; **nanotechnology** is the engineering use of those phenomena. One nanometre is 10^{-9} m. A material needs only *one* dimension in the nano range to earn the label, which gives the standard family:

- **Nanoparticles** (0-dimensional) — all three dimensions small: nano silver, nano TiO_2.
- **Nanofibres and nanotubes** (1-dimensional) — diameter small, length large: electrospun fibres, carbon nanotubes.
- **Nanocoatings and nanoplates** (2-dimensional) — thickness small: sol–gel layers, nanoclay platelets, graphene flakes.

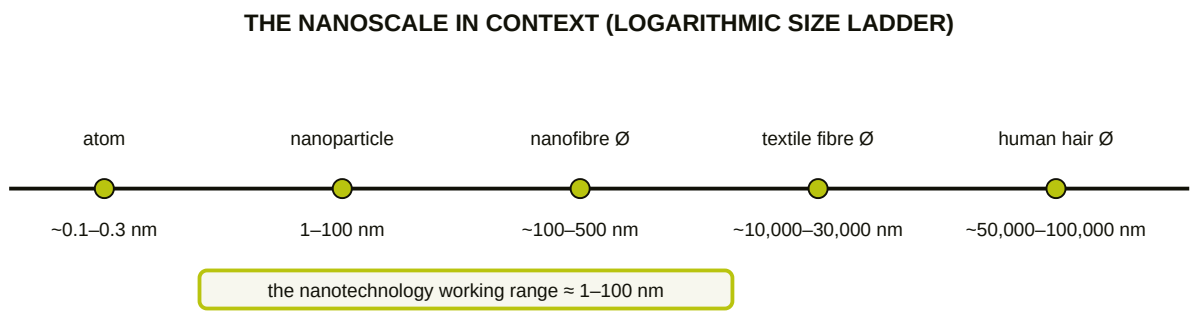


Figure 1 — Size ladder: the nanoscale sits between atoms and ordinary textile fibres — roughly a thousand times finer than the fibres of earlier topics.

2. The surface effect — the first reason small is different

For a sphere of radius r, SA/V = 3/r (derived in the YM-4203 nanotechnology sheet and used again in Topic 8). Consequences at textile scale:

- A particle of radius 1 μm has SA/V = 3×10^6 m^{-1}; at radius 10 nm, SA/V = 3×10^8 m^{-1} — one hundred times more surface per unit volume.
- The **fraction of atoms at the surface** explodes. For a sphere of diameter d built of atoms of size a ≈ 0.3 nm, the surface fraction is roughly 6a/d: about **36% at d = 5 nm, 18% at 10 nm, 1.8% at 100 nm**, and below 0.2% at 1 μm. Surface atoms have unsatisfied bonds — they are where chemistry, adhesion, catalysis and melting all start.

3. Size dependence of properties (the syllabus list)

3.1 Mechanical

At small sizes, defect populations shrink: a nanofibre or whisker contains fewer of the flaws that start failure, so measured strengths approach theoretical values — one reason nanofibres and carbon nanotubes report such high specific strengths. Conversely, nanoparticle-filled polymers stiffen at very low loadings because the interface (load-transfer area) per unit volume is enormous — the same surface arithmetic as Section 2, applied to mechanics.

3.2 Thermodynamic

High surface energy makes small particles melt lower and sinter easier. The melting point of a nanoparticle falls as its radius shrinks (surface atoms are less tightly held), which is why silver nanoparticles can be sintered into conductive tracks at temperatures a fabric survives, while bulk silver melts at 962 °C. Solubility also rises with curvature — small particles dissolve faster, the working principle behind nano-silver's ion release.

3.3 Electronic and optical

When a semiconductor particle becomes smaller than the natural size of its electron–hole pair (the exciton), its band gap *widens* — **quantum confinement**. The colour of light absorbed and emitted then depends on particle size: the same material, CdSe for example, can be tuned across the visible spectrum by size alone. Photon energy follows E = hc/λ, conveniently E(eV) ≈ 1240/λ(nm): 400 nm light carries 3.10 eV, 700 nm carries 1.77 eV. Bulk CdSe has a band gap of about 1.74 eV (absorption edge near 712 nm, deep red); shrinking the crystal pushes the edge toward the blue — a property that cannot exist at bulk scale. Plasmonic effects in metal nanoparticles (silver, gold) similarly give strong, size-dependent colour and the surface-enhanced sensing used in some smart-textile research.

3.4 Catalytic

Catalysis happens at surfaces, so catalytic activity per gram tracks surface area: nanoparticulate TiO_2 is photocatalytic enough to drive the self-cleaning finishes of Topic 9, and nano-sized catalyst particles expose a far larger share of their (often precious) metal atoms to the reactants. The shadow side is the

same fact: high reactivity is indifferent to our intentions, so nano-catalysts and antimicrobials must be placed and bound deliberately (safety discussion in the YM-4203 sheet and Topic 9).

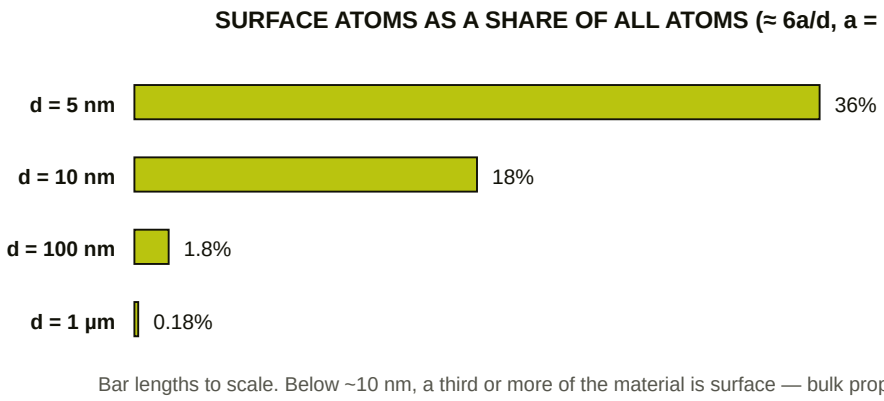


Figure 2 — Surface-atom fraction versus particle diameter. This single curve explains most of nanoscience.

4. Top-down and bottom-up

Two philosophies make nanostructures. **Top-down** starts with bulk material and cuts it down: milling, grinding, lithography, melt-blowing toward finer fibres. It is robust and industrial but fights physics as sizes shrink, and milled particles have wide size distributions. **Bottom-up** builds from atoms and molecules: chemical synthesis of nanoparticles (Topic 7), sol–gel coating, electrospinning from solution, layer-by-layer assembly (Topic 9). Bottom-up gives the narrow distributions and designed surfaces the applications need — textiles are overwhelmingly a bottom-up field, because our nanostructures are usually applied from liquids onto fibres. A third route specific to fibres deserves its name: **phase-separation spinning** (sea-island bicomponent fibres split or dissolved into micro- and nano-fibrils) reaches nanofibre fineness on modified conventional machinery.

5. Measurement belongs to the definition

A nanometre cannot be seen in an optical microscope. The field therefore leans on the characterisation toolkit of Topic 7 — electron microscopy (SEM/TEM), atomic force microscopy (AFM), X-ray diffraction (XRD) and light scattering (DLS). A disciplined habit for this whole course: *any statement about a nanomaterial should name the technique that measured it.*

Glossary

Nanometre — 10^{-9} m. **Quantum confinement** — widening of a semiconductor's band gap when the crystal is smaller than its exciton. **Exciton** — a bound electron–hole pair. **Plasmon** — collective oscillation of electrons at a metal surface, giving strong size-dependent optical effects. **Sintering** —

bonding of particles into a solid mass below the bulk melting point. **Top-down / bottom-up** — making nanostructures by subdivision of bulk material / by assembly from atoms and molecules.

Key points

• Nanoscience works at 1–100 nm in at least one dimension: particles (0D), fibres/tubes (1D), coatings/plates (2D).
• Surface fraction ≈ 6a/d: at 5 nm over a third of the atoms are surface atoms — reactivity, melting, solubility and catalysis all follow.
• Mechanical properties shift because small volumes hold fewer defects and interfaces dominate; electronic properties shift by quantum confinement ($E \approx 1240/\lambda$ eV).
• Textile nanotechnology is mostly bottom-up: synthesis, coating and spinning from liquids onto fibres.
• Every nano claim needs a measurement technique behind it — SEM/TEM, AFM, XRD, DLS (Topic 7).

Practice

1. Estimate the surface-atom fraction for particles of diameter 20 nm and 2 nm (a = 0.3 nm), and state one practical consequence of the 2 nm value.
2. Using $E \approx 1240/\lambda$, find the photon energy at 550 nm and the wavelength corresponding to a band gap of 2.48 eV. What does quantum confinement do to the absorption edge of a semiconductor as its crystals shrink?
3. Classify as top-down or bottom-up, with one line of justification: (a) ball-milling pigment to sub-micrometre size; (b) growing silver nanoparticles by chemical reduction in solution; (c) producing nanofibres by electrospinning.

Answer key

1. 20 nm: $6 \times 0.3/20$ = **9%**. 2 nm: $6 \times 0.3/2$ = **90%** — nearly the whole particle is surface, so it behaves almost like a giant molecule: extremely reactive, unstable against growth/agglomeration, and its “bulk” properties (melting point, band structure) barely apply.
2. E(550 nm) = 1240/550 ≈ **2.25 eV**. λ for 2.48 eV = 1240/2.48 = **500 nm**. Confinement *widens* the band gap, so the absorption edge moves to **shorter wavelength (blue shift)** as crystals shrink.
3. (a) **Top-down** — bulk material subdivided by mechanical force. (b) **Bottom-up** — particles assembled atom by atom from dissolved ions. (c) **Bottom-up** — fibres formed by drawing a polymer solution into nanoscale filaments (assembly from a molecular solution, not cutting of a bulk fibre).